

HMMT Invitational Competition

April 2016

1. [5] Theseus starts at the point $(0, 0)$ in the plane. If Theseus is standing at the point (x, y) in the plane, he can step one unit to the north to point $(x, y + 1)$, one unit to the west to point $(x - 1, y)$, one unit to the south to point $(x, y - 1)$, or one unit to the east to point $(x + 1, y)$. After a sequence of more than two such moves, starting with a step one unit to the south (to point $(0, -1)$), Theseus finds himself back at the point $(0, 0)$. He never visited any point other than $(0, 0)$ more than once, and never visited the point $(0, 0)$ except at the start and end of this sequence of moves.

Let X be the number of times that Theseus took a step one unit to the north, and then a step one unit to the west immediately afterward. Let Y be the number of times that Theseus took a step one unit to the west, and then a step one unit to the north immediately afterward. Prove that $|X - Y| = 1$.

Proposed by: Mitchell Lee

The path Theseus traces out is a closed, non-self-intersecting path in the plane. Since each move is along a line segment, the path forms the boundary of a polygon in the plane. WLOG this polygon is traversed counterclockwise (i.e. with the interior of the polygon always on Theseus's left); the argument in the clockwise case is analogous, with a sign flip.

Let $i = 0$ correspond to the direction east, $i = 1$ correspond to north, and so forth. For $0 \leq i \leq 3$ and $j = i \pm 1$ let X_{ij} be the number of times Theseus switches from going in direction i to going in direction j (where indices are taken mod 4). So, $X = X_{1,2}, Y = X_{2,1}$. We have the following relations among the X_{ij} :

The sum $n = \sum X_{i,j}$ is the number of vertices of the polygon. The sum of the interior angles of the polygon is thus $\pi(n-2)$. Each counterclockwise turn contributes $\frac{\pi}{2}$, and each clockwise turn contributes $\frac{3\pi}{2}$ to this sum. The number of the former is counted by $\sum X_{i,i+1}$, and the latter by $\sum X_{i+1,i}$. So, $\pi(n-2) = \frac{\pi}{2} \sum X_{i,i+1} + \frac{3\pi}{2} \sum X_{i+1,i} = \pi n + \frac{\pi}{2} \sum (X_{i+1,i} - X_{i,i+1})$, which gives $\sum (X_{i,i+1} - X_{i+1,i}) = 4$.

Finally, observe by starting from the middle of any edge that since we start and end in the same direction, the number of times we turn into direction i must equal the number of times we turn out of it. So $X_{i,i+1} + X_{i,i-1} = X_{i+1,i} + X_{i-1,i}$, which rearranges to give $X_{i,i+1} - X_{i+1,i} = X_{i-1,i} - X_{i-1,i-1}$. So $4 = \sum (X_{i,i+1} - X_{i+1,i}) = 4(X_{1,2} - X_{2,1}) = 4(X - Y)$.

So $X - Y = 1$ as desired.

2. [7] Let ABC be an acute triangle with circumcenter O , orthocenter H , and circumcircle Ω . Let M be the midpoint of AH and N the midpoint of BH . Assume the points M, N, O, H are distinct and lie on a circle ω . Prove that the circles ω and Ω are internally tangent to each other.

Proposed by: Evan Chen

Let R be the circumradius of $\triangle ABC$. Recall that the circumcircle of $\triangle HBC$ has radius equal to R . By a homothety at H with factor 2 at H , it follows that $\triangle HMN$ has circumradius $R/2$. Thus ω has circumradius $R/2$, and since O lies on ω the conclusion follows.

Remark. This problem is a slight modification of a proposal by *Dhroova Aiylam*.

3. [8] Denote by $\mathbb{N}$ the positive integers. Let $f : \mathbb{N} \rightarrow \mathbb{N}$ be a function such that, for any $w, x, y, z \in \mathbb{N}$,

$$f(f(f(z)))f(wxf(yf(z))) = z^2 f(xf(y))f(w).$$

Show that $f(n!) \geq n!$ for every positive integer n .

Proposed by: Pakawut Jiradilok

If $f(z_1) = f(z_2)$, then plugging in (w, x, y, z_1) and (w, x, y, z_2) yields $z_1 = z_2$. Thus, f is injective. Substitution of (z, w) with $(1, f(f(1)))$ in the main equation yields

$$f\left(f(f(1))xf(yf(1))\right) = f(xf(y))$$

Because f is injective, we have

$$f(yf(1)) \cdot f(f(1)) = f(y), \forall y \in \mathbb{N} \quad (*)$$

We can easily show by induction that

$$f(y) = f(yf(1)^n)f(f(1))^n, \forall n \in \mathbb{N}$$

Thus, $f(f(1))^n | f(y), \forall n \in \mathbb{N}$ which implies that $f(f(1)) = 1$. Using this equality in $(*)$ yields $f(1) = 1$. Substitution of (y, z) with $(1, 1)$ in the main equation yields $f(xy) = f(x)f(y)$.

So,

$$f(n!) = f\left(\prod_{i=1}^n i\right) = \prod_{i=1}^n f(i).$$

By injectivity, each $f(i)$ in this product is a distinct positive integer, so their product is at least $\prod_{i=1}^n i = n!$, as desired.

Remark 1. The equation condition of f in the problem is actually equivalent to the following two conditions combined:

$$(C1) \ f(xy) = f(x)f(y), \forall x, y \in \mathbb{N}, \text{ and}$$

$$(C2) \ f(f(f(x))) = x, \forall x \in \mathbb{N}$$

4. [10] Let P be an odd-degree integer-coefficient polynomial. Suppose that $xP(x) = yP(y)$ for infinitely many pairs x, y of integers with $x \neq y$. Prove that the equation $P(x) = 0$ has an integer root.

Proposed by: Victor Wang

Proof. Let n be the (odd) degree of P . Suppose, for contradiction, that P has no integer roots, and let $Q(x) = xP(x) = a_n x^{n+1} + a_{n-1} x^n + \dots + a_0 x^1$. WLOG $a_n > 0$, so there exists $M > 0$ such that $Q(M) < Q(M+1) < \dots$ and $Q(-M) < Q(-M-1) < \dots$ (as $n+1$ is even).

There exist finitely many solutions (t, y) for any fixed t , so infinitely many solutions (x, y) with $|x|, |y| \geq M$. By definition of M , there WLOG exist infinitely many solutions with $|x| \geq M$ and $y \geq -x > 0$ (otherwise replace $P(t)$ with $P(-t)$).

Equivalently, there exist infinitely many (t, k) with $t \geq M$ and $k \geq 0$ such that

$$0 = Q(t+k) - Q(-t) = a_n[(t+k)^{n+1} - (-t)^{n+1}] + \dots + a_0[(t+k)^1 - (-t)^1].$$

If $k \neq 0$, then $Q(x+k) - Q(-x)$ has constant term $Q(k) - Q(0) = kP(k) \neq 0$, and if $k = 0$, then $Q(x+k) - Q(-x) = 2(a_0 x^1 + a_2 x^3 + \dots + a_{n-1} x^n)$ has x^1 coefficient nonzero (otherwise $P(0) = 0$), so no fixed k yields infinitely many solutions.

Hence for any $N \geq 0$, there exist $k = k_N \geq N$ and $t = t_N \geq M$ such that $0 = Q(t+k) - Q(-t)$. But then the triangle inequality yields

$$\underbrace{a_n[(t+k)^{n+1} - (-t)^{n+1}]}_{=\sum_{i=0}^{n-1} -a_i[(t+k)^{i+1} - (-t)^i]} \leq 2(t+k)^n(|a_{n-1}| + \dots + |a_0|). \quad (1)$$

But $(t+k)^{n+1} - t^{n+1} \geq (t+k)^{n+1} - t \cdot (t+k)^n = k \cdot (t+k)^n$,¹ so $a_n k_N \leq 2(|a_{n-1}| + \dots + |a_0|)$ for all N . Since $k_N \geq N$, this gives a contradiction for sufficiently large N . $\square$

¹However, the mean value theorem estimate $(t+k)^{n+1} - t^{n+1} \geq k \cdot (n+1)t^n$ will not suffice a priori, if k happens to be much larger than t . Of course, it is easy to prove that k cannot be too much larger than t (even before finishing the problem); the execution would just take slightly longer.

Remark 2. Another way to finish after the triangle inequality estimate (1) is to observe that $(t+k)^{n+1} - (-t)^{n+1} = (t+k)^{n+1} - t^{n+1} \geq t^n \cdot k + k^{n+1}$ (by binomial expansion), so $\frac{k_N(t_N^n + k_N^n)}{(t_N + k_N)^n}$ (which by Holder's inequality is at least $\frac{k_N}{2^{n-1}} \geq \frac{N}{2^{n-1}}$) is bounded above (independently of N), contradiction. (Here we use Holder's inequality to deduce $(1+1)^{n-1}(a^n + b^n) \geq (a+b)^n$ for positive reals a, b .)

Remark 3. The statement is vacuous for even-degree polynomials. The case of degree 3 polynomials was given as ISL 2002 A3.

5. [12] Let $S = \{a_1, \dots, a_n\}$ be a finite set of positive integers of size $n \geq 1$, and let T be the set of all positive integers that can be expressed as sums of perfect powers (including 1) of distinct numbers in S , meaning

$$T = \left\{ \sum_{i=1}^n a_i^{e_i} \mid e_1, e_2, \dots, e_n \geq 0 \right\}.$$

Show that there is a positive integer N (only depending on n) such that T contains no arithmetic progression of length N .

Proposed by: Yang Liu

Answer: N/A

In general we can assume that each $a_i > 1$, since replacing $a_i = 1$ by some large integer a creates a set T containing the original T as a subset (by setting $e_i = 0$).

We proceed by induction on n . For the base case $n = 1$, an arithmetic progression of length at least 3 would give $a_1^{e_1} + a_1^{e_3} = 2a_1^{e_2}$ where $e_3 > e_2 > e_1$. But $a_1^{e_3} \geq 2a_1^{e_2}$, so this is impossible. Thus our result holds for $n = 1$.

Assume the result is true for some $n - 1$, and let A_{n-1} be a number such that the longest progression when $|S| = n - 1$ has length less than A_{n-1} . Let M be a large integer that we will choose later. Take an arithmetic progression of length $2M + 1$, calling the terms $b_1, b_2, \dots, b_{2M+1}$. Note that $b_{2M+1} \leq 2b_{M+1}$, since b is a sequence of positive integers. For each term from b_{M+1} to b_{2M+1} assign to it the maximum power that is part of the sum. Call this value $c(b_i)$. More explicitly, if $b_i = \sum_{j=1}^n a_j^{e_j}$, then $c(b_i) = \max(a_1^{e_1}, a_2^{e_2}, \dots, a_n^{e_n})$.

Since $\frac{b_{M+1}}{n} \leq c(b_i) \leq b_{2M+1} \leq 2b_{M+1}$ for $M+1 \leq i \leq 2M+1$, there are at most $\sum_{j=1}^n (\log_{a_i} 2n+1) \leq n(\log_2 2n+1)$ different values of $c(b_i)$. By Van der Waerden's Theorem, there exists a value of M such that coloring an arithmetic progression of length M with $n(\log_2 2n+1)$ colors yields a monochromatic arithmetic progression of length A_{n-1} . In particular, we can take $M = W(A_{n-1}, n(\log_2 2n+1))$, where $W(n, k)$ denotes the Van der Waerden number. So, we color $b_{M+1}, \dots, b_{2M+1}$ by their $c(b_i)$. Subtracting the common perfect power from each term of the monochromatic arithmetic progression obtained gives an arithmetic progression of A_{n-1} integers expressible as the sum of perfect powers of distinct numbers in $S \setminus a_j$. By the inductive hypothesis, this is a contradiction. So no arithmetic progression of length $2M + 1$ can be contained in T , and we can take $A_n = 2M + 1$.

By induction, we thus have such an A_n for all n .